

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

# Chapter 16: Loci (Locus and its Constructions)

Previous year board questions, topic-wise: 5 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

## 1 Loci in a triangle

- Statement 1: The point equidistant from three non-collinear points  $D$ ,  $E$  and  $F$  is the circumcentre of  $\triangle DEF$ . Statement 2: The incentre of a triangle is the point where the bisectors of its angles meet. (a) both true (b) both false (c) 1 true, 2 false (d) 1 false, 2 true

ICSE 2025, 1 mark

- The circumcentre of a triangle is the point that is: (a) equidistant from the three sides (b) equidistant from the three vertices (c) the meeting point of the three medians (d) the meeting point of the three altitudes

ICSE 2024, 1 mark

## 2 Construction questions

- Using ruler and compasses, construct  $\angle ABC = 90^\circ$  with  $AB = 6$  cm and  $BC = 8$  cm. Construct (a) the locus of points equidistant from  $B$  and  $C$  (b) the locus of points equidistant from  $A$  and  $B$  (c) the locus of points at the fixed distance  $OA$  from  $O$ , where  $O$  is the point common to (a) and (b) (d) the locus of points equidistant from  $BA$  and  $BC$ .

ICSE 2024, 4 marks

- Using ruler and compasses, draw a circle of radius 4.5 cm and a chord  $AB = 6$  cm. (a) Construct the locus of points equidistant from  $A$  and  $B$ , and mark the point  $D$  where it meets the circle. (b) Join  $AD$ , construct the locus of points equidistant from  $AD$  and  $AB$ , and mark the point  $C$  where it meets the circle. (c) Join  $BC$  and  $CD$  and measure  $CD$ .

ICSE 2020, 4 marks

- Using ruler and compasses, construct a semicircle on the diameter  $BC = 7$  cm. Locate the point  $A$  on the semicircle that is equidistant from  $B$  and  $C$ . Complete the cyclic quadrilateral  $ABCD$  with  $D$  equidistant from  $AB$  and  $BC$ . Measure  $\angle ADC$ .

ICSE 2019, 4 marks

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